

HARMONY: Passive Fins Everywhere

—A Net-Zero Passive Retrofit Framework^[1]

Summary

"The best way to predict the future is to create it." — Peter Drucker

Passive solar shading cannot be sized only at solar-noon on solstices: campuses must manage hourly heat gain, glare, and future warming. Sungrove (low-latitude) faces high irradiance, heat waves, and a net-zero cooling target by 2040; Borealis (high-latitude) is heating-dominated and depends on effective thermal storage. Both also require a Student Union prototype that performs throughout the day, under non-cardinal orientations, and under projected climate conditions.

We propose *HARMONY* (Hourly Academic Retrofit Model with Operable shading and energy balance), an hourly thermal–visual–energy framework that evaluates passive retrofits and new-build strategies on *occupied academic hours*. By coupling solar geometry, shading physics, comfort constraints, glare-hours, daylight penalties, and PV offset, *HARMONY* produces constructible parameter sets and defensible tradeoffs rather than “over-shade” heuristics.

Model I: HARMONY-NetZero (Sungrove retrofit). We co-optimize glazing (SHGC, U , VLT), south overhangs, east/west fins, night ventilation, added thermal mass, and operable interior shading. The selected package reduces *academic cooling* by **44%** and *annual cooling* by **48%** while meeting comfort and glare limits. Net-zero feasibility is strong, and only **162.5 m²** of PV is required for full offset.

Model II: HARMONY-NZ-ThermalMass (Borealis retrofit). We extend the balance with an explicit effective capacitance (thermal-mass scaling) and seasonal shade logic. The best feasible design cuts *academic heating* from **49,798.8** to **40,119.2 kWh (19.4%)**, and reduces glare-hours from 394 to 233 (below the 275.8 target) with acceptable daylight deficit.

Model III: HARMONY-NZ-TM-Generalized (multi-city transfer). Across six climates, feasible/robust solutions maintain comfort and glare control while keeping roof-PV requirements on the order of **10² m²** (typically **~103–240 m²**), showing the design logic is transferable.

Model IV: HARMONY’s Application (Student Union new-build). We optimize rotation and facade-specific WWR together with passive envelope and shading. A passive “HVAC-OFF” stress test exposes outage risk (**peak 64.3°C**; **5,195 comfort-violation hours**), so we define a minimal “backup HVAC” mode: electrical cooling drops by **≈21%** with **0 glare-hours**. Worst-case 2040 raises cooling electricity by approximately **15%**.

We test robustness under 2040 ΔT , heatwave operations, and outage survivability, and run global sensitivity via PRCC with bootstrap confidence intervals. Results highlight interpretable, actionable levers (SHGC, fin geometry, night ventilation, thermal mass, WWR) and support a best-feasible and robust short-list.

Keywords: passive solar shading; academic-hour occupancy; brise-soleil fins; thermal mass; net-zero cooling; PRCC.

Contents

1	Introduction	2
1.1	Background	2
1.2	Restatement of the Problem	2
1.3	Our Work	3
2	Assumptions	4
3	Notations	4
4	Model Preparation	6
4.1	Data Overview	6
4.2	Data Collection	7
4.3	Data Preparation	7
4.4	Mathematical Modeling of Solar Geometry and Facade Irradiance	8
5	HARMONY-NetZero: Low-Latitude	9
5.1	Why We Use HARMONY	9
5.2	Modeling Framework	9
5.3	Parameter Search and Selected Retrofit Package	11
5.4	Instantiated Inputs and Key Results	12
5.5	Robustness, Sensitivity, and 2040 Prediction	12
6	HARMONY-NZ-ThermalMass: High-Latitude	13
6.1	Model Advantages Compared to Traditional Approaches	14
6.2	Model Formulation	14
6.3	Model Results, Conclusions, and Interpretation	15
6.4	Future Robustness ^[2]	17
6.5	Core Clarifications (Avoiding Common Misinterpretations)	17
7	HARMONY-NZ-TM-Generalized: Multiple Locations	17
7.1	Advantages Over Traditional Models	17
7.2	Model Formulas and Description	18
7.3	Model Results, Conclusions, and Explanations	18
7.4	Core Clarifications	19
8	HARMONY's Application	20
8.1	HARMONY's Advantages	20
8.2	Model Formulation	21
8.3	Results, Conclusions, and Explanations	21
8.4	Passive-first comparison (HVAC OFF) and backup HVAC	22
8.5	Core Clarifications	23
9	Weaknesses and Improvements	24
9.1	Weaknesses	24
9.2	Improvements	24
	References	24
	Letter to Sungrove University	25

1 Introduction

1.1 Background

Against the backdrop of global warming and increasingly frequent extreme heat events, building energy consumption—particularly the rapid rise in summer cooling loads—has become a key challenge constraining campus sustainability. University campuses located in low-latitude, high-radiation zones often face the dual predicament of intense solar radiation, glare interference, and high cooling energy costs. Traditional solutions relying solely on mechanical cooling not only consume substantial energy but also conflict with long-term carbon reduction goals. Consequently, leveraging the predictability of the sun's path through passive solar shading strategies—reducing summer heat gain while preserving beneficial winter daylight—has emerged as a critical research topic in architectural design and engineering decision-making for both building retrofits and new construction.^[3]

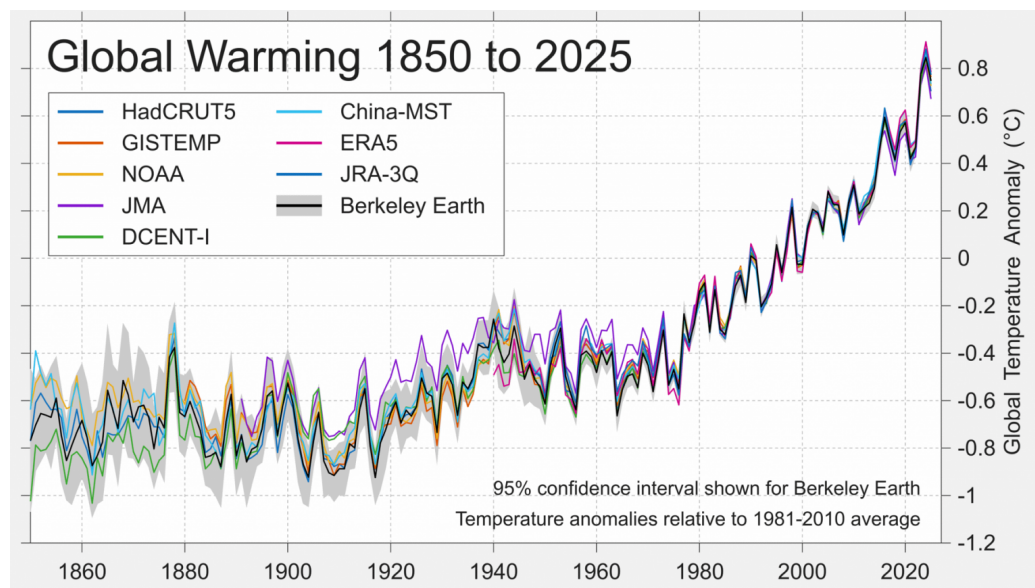


Figure 1: the changes of global climate over past few years

*Global Temperature Report for 2025, <https://berkeleyearth.org/global-temperature-report-for-2025/> by Robert Rohde, used under CC BY 4.0 / Cropped from original

1.2 Restatement of the Problem

Within this context, COMAP formulates a campus-building retrofit problem represented by two notional institutions: Sungrove University and Borealis University. Our work is organized around the following objectives:

1. Develop a modeling-based approach to mitigate glare during instructional hours in low-latitude regions, while reducing cooling demand/costs and maintaining acceptable indoor comfort conditions.
2. Extend the model from Task 1 to high-latitude regions to address insufficient thermal availability under very low ambient temperatures, and to make effective use of thermal storage.

3. Calibrate and refine the model to improve its applicability and transferability across cities with distinct climate regimes.
4. Analyze environmental characteristics across different latitude bands and, based on these findings, design passive solar-shading devices suitable for deployment at a notional university campus.

1.3 Our Work

We structure the full solution as a modular pipeline that directly maps to COMAP's four technical requirements and the final policy letter deliverable.

1. **Data and operating context.** We ingest hourly EPW weather and construct an academic occupancy mask to ensure all KPIs (cooling/heating energy, glare-hours, comfort violations) are computed on *teaching-relevant* hours.
2. **Hourly solar-shading engine.** We compute solar position and facade incidence each hour, then evaluate geometric shading from south overhangs and east/west vertical fins; interior shades are modeled as *operable*, activated only when solar/glare thresholds are exceeded.
3. **Coupled performance metrics.** We link solar gains to a one-zone thermal balance (with HVAC accounting when enabled), compute occupied-hour comfort-violation hours, and track glare-hours and daylight-penalty impacts to avoid over-shading that forces excessive lighting energy.
4. **Design-space exploration and optimization.** For each requirement, we screen candidate bundles over glazing, shading geometry, thermal-mass scale, and night ventilation, selecting feasible designs that minimize energy while meeting glare/comfort constraints, then reporting constructible parameter sets and cost proxies.
5. **Robustness (future + extremes) and interpretability.** We re-evaluate selected designs under 2040 warming deltas, heatwave operations, and outage survivability, and run PRCC sensitivity (with bootstrap CIs) to rank which inputs most drive energy and robustness. This produces both a *best-feasible* recommendation and a *robust* short-list suitable for campus standards.

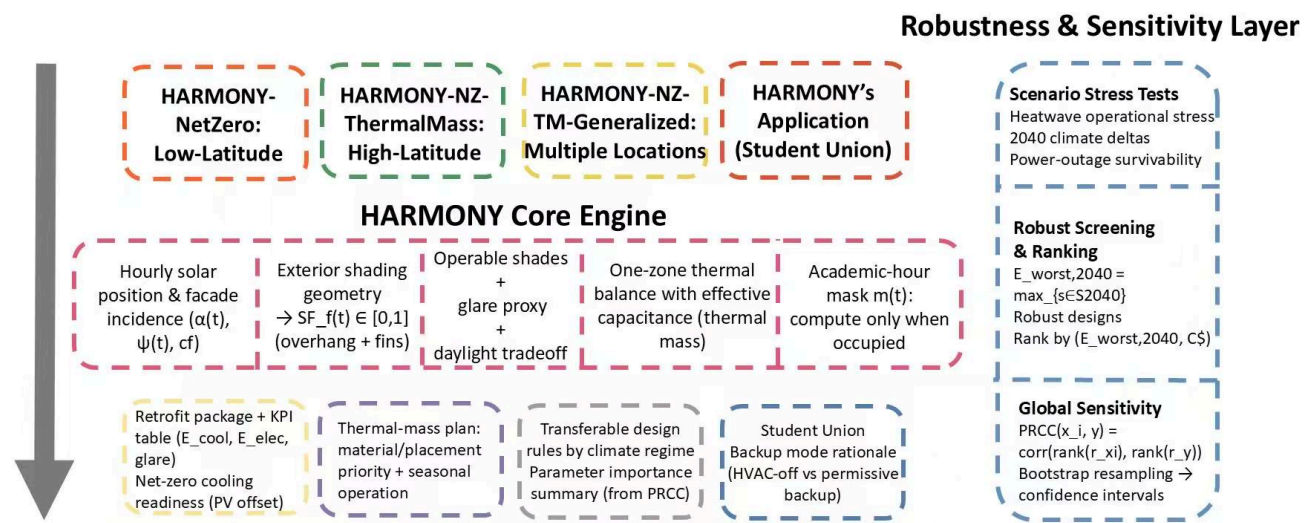


Figure 2: Our work

2 Assumptions

1. **Representative city selection.** Sungrove University is assumed to be located in Phoenix, and Borealis University is assumed to be located in Fairbanks.
Rationale. These two cities are consistent with the problem's notional setting, and they also provide representative, economically developed, and data-rich contexts that enhance the practical relevance of campus-building modeling.
2. **Storey heights.** The classroom storey height is assumed to be 3.2 m, and the student activity center storey height is assumed to be 4.0 m.
Rationale. According to *Architectural Graphic Standards*, typical classroom clear heights are approximately 9–12 ft (2.74–3.66 m)^[2]. We therefore set the classroom height to 3.2 m. Given the multi-functional program requirements of a student activity center, we adopt a larger height of 4.0 m.
3. **Climate data source.** Local airport weather records are used as the reference climate data for each campus.
Rationale. Airport datasets are among the most accurate and readily available site-specific meteorological records. At the city scale, spatial variations in air temperature at the same time are assumed negligible for the purposes of this study.
4. **Single-zone, hourly thermal model.** The building's dominant thermal dynamics are approximated using a single-zone model with an hourly time step (single-zone, time-marching), which is sufficient to support comparative evaluation of shading and thermal-mass strategies.
Rationale. The building is treated as a single, well-defined volume with relatively simple programmatic subdivision. Adopting a single-zone representation reduces computational burden while providing a consistent baseline for subsequent refinement toward more detailed time-step simulations.
5. **Exterior shading prioritized; fixed-adjustable combinations allowed.** Exterior shading is prioritized, and combined configurations of *fixed* shading and *adjustable* shading (scheduled by season and/or time of day) are permitted.
Rationale. Exterior shading intercepts solar radiation before it enters the interior, thereby reducing solar heat gains more effectively than interior shading.
6. **No surrounding obstructions.** The building is assumed to have no surrounding obstructions (e.g., adjacent buildings, trees, or terrain features) that cast shadows on the façades.
Rationale. To simplify computation, we assume unobstructed solar exposure and neglect the impact of external context on incident solar radiation in the modeling.
7. **Class-hour schedule references.** The instructional-time schedule is referenced from Arizona State University (ASU) for the low-latitude case and the University of Alaska Fairbanks (UAF) for the high-latitude case. *Rationale.* These schedules are representative of the assumed locations and provide realistic proxies for defining occupied teaching periods.

3 Notations

Symbols	Description	Unit
t	hour index	h
$T_a(t)$	outdoor dry-bulb temperature	°C
$\alpha(t), \gamma(t)$	solar altitude angle / azimuth angle	rad or deg
$A_{w,f}$	window area oriented $f \in \{S, N, E, W\}$	m ²
UA	overall building heat transfer coefficient	W/K
C_{th}	indoor equivalent thermal capacitance	J/K
$T_z(t)$	indoor zone temperature	°C
$SHGC$	solar heat gain coefficient of glazing	–
VLT	visible light transmittance of glazing	–
U_{win}	window heat transfer coefficient	W/m ² K
D_{oh}	south-facing overhang depth	m
$D_{fin,E/W}$	depth of fins for east/west orientations	m
$S_{fin,E/W}$	shading factor for east/west orientations	–
$SF_f(t)$	shading reduction factor for orientation f	–
$Q_{sol}(t)$	solar heat gain through windows	W
$Q_{int}(t)$	internal heat gain from occupants, equipment, and lighting	W
$Q_{vent}(t)$	sensible heat exchange due to ventilation	W
$Q_{cool}(t)$	cooling power required to maintain set temperature	W
E_{cool}	annual cooling energy (integral of Q_{cool})	kWh
COP	coefficient of performance for cooling	–
$u(t)$	curtain state	–
$E_{light,add}$	additional lighting electricity consumption due to curtain closing	kWh
$E_{tot,elec}$	total electricity consumption	kWh
EPV	annual photovoltaic generation on roof	kWh
$T_{in}(t)$	Indoor air temperature at time t	°C
$T_{out}(t)$	Outdoor air temperature at time t	°C
$F_{sh}(t)$	Shading transmittance factor (0–1); 1 denotes no shading	–
$\dot{m}c_p$	Equivalent heat-capacity flow rate due to ventilation/infiltration heat exchange	W/K
$C_{eff}^{[5]}$	Equivalent indoor thermal capacitance	J/K
E_{heat}	Heating energy use / heating demand over the evaluation period	kWh

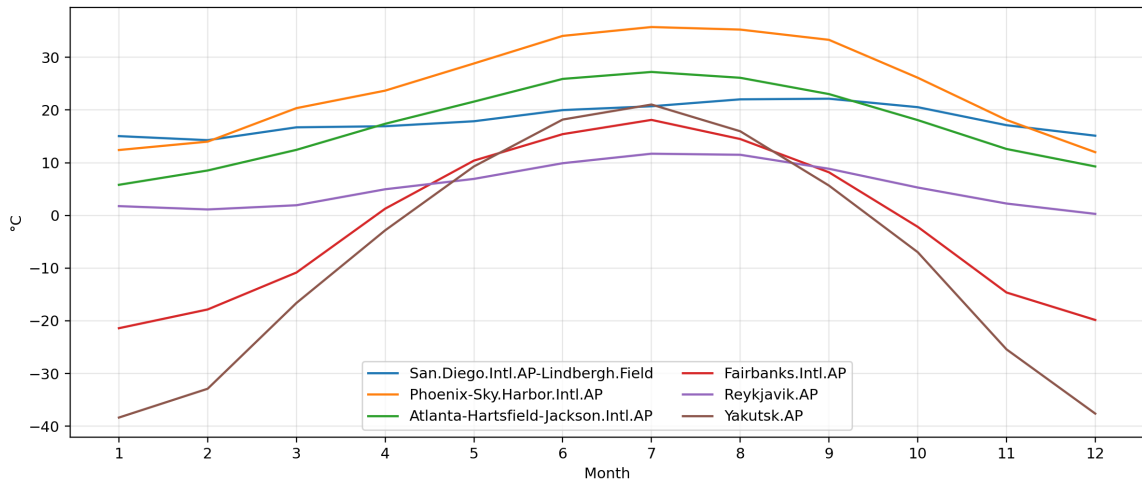
where we define the main parameters while specific value of those parameters will be given later.

4 Model Preparation

4.1 Data Overview

Because the prompt does not provide meteorological inputs or class schedules for Sungrove University and Borealis University, we use the data extracted from the hypothetical scenarios and extended to other different cities for Model I-III.

For each city, we collect hourly weather records from 2009 to 2023, including global horizontal irradiance (GHI), direct normal irradiance (DNI), diffuse horizontal irradiance (DHI), and dry-bulb air temperature (T_a). We also compile academic calendar and class-hour schedules from Arizona State University (ASU) and the University of Alaska Fairbanks (UAF), including semester start dates and typical teaching-time windows. Due to the large data volume, we report some of these inputs via visualizations rather than listing full time series.



(a) Monthly mean temperature

Figure 3: Temperature contrasts across representative cities.

4.2 Data Collection

Table 1: Primary data sources used for model inputs and validation.

Database Names	Data category	Database Websites
Station observations	Meteorology	https://www.ncei.noaa.gov/products/land-based-station/integrated-surface-database
EPW weather files	Meteorology	https://climate.onebuilding.org/WMO_Region_4_North_and_Central_America/USA_United_States_of_America/index.html
Instructional periods(ASU)	Academic schedule	https://registrar.asu.edu/registrar-semester-calendar
Instructional periods(UAF)	Academic schedule	https://www.uaf.edu/catalog/archives.php
Glass properties	Materials	https://sciglass.uni-jena.de/
Engineering material properties	Materials	https://www.matweb.com/

4.3 Data Preparation

4.3.1 Data Extraction

- **Meteorology**

From each EPW record (after line 8), we extract the following fields (0-based column indices):

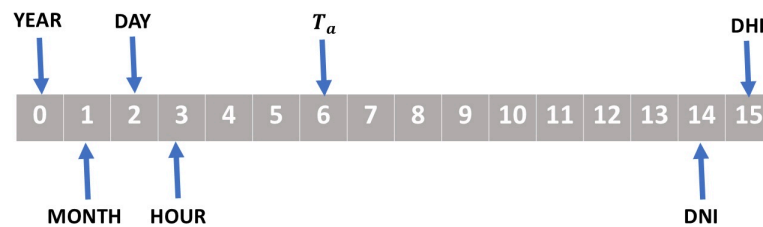


Figure 4: the simulated diagram of the buildings

- **Academic schedule**

Semester start/end dates and typical teaching-time windows are obtained from official university web pages. These dates are manually encoded in the model to construct the occupancy/class-hour mask used in subsequent energy and glare calculations.

- **Materials**

Based on the optimized design parameters produced by the model (e.g., glazing properties and shading geometry), we query online building-material catalogs and datasheets to identify commercially available products whose specifications most closely match the selected parameters.

The matched products are then treated as a feasible construction specification for the proposed retrofit.

4.3.2 Data Cleaning

To improve numerical robustness, we apply basic sanity checks and gap-filling prior to simulation.

Dry-bulb temperature T_a

- Physically implausible values (e.g., $|T_a| > 80^\circ\text{C}$) are treated as missing and set to NaN.
- Missing values (NaN) are imputed using linear interpolation to avoid numerical failures or unrealistic discontinuities in subsequent energy/comfort calculations.

4.4 Mathematical Modeling of Solar Geometry and Facade Irradiance

To accurately evaluate the thermal load and glare potential, we establish a rigorous solar trajectory model. The calculation proceeds from celestial mechanics to local coordinates, and finally to vector projection on building surfaces.

Step A: Solar Position Calculation

The model inputs include latitude ϕ , longitude λ , time zone z , day of year n , and local clock time h . Note that we define the calculation timestamp as the hour midpoint ($h = t\%24 + 0.5$).

First, the solar declination angle δ is determined by Cooper's approximation:

$$\delta = 23.45^\circ \sin\left(\frac{360^\circ(284 + n)}{365}\right) \quad (1)$$

To correct for orbital eccentricity, we calculate the auxiliary angle B and the Equation of Time (EoT):

$$B = \frac{360^\circ(n - 81)}{364}, \quad EoT = 9.87 \sin(2B) - 7.53 \cos(B) - 1.5 \sin(B) \quad (2)$$

We convert local clock time to Local Solar Time (LST) using the Local Standard Time Meridian ($LSTM$) and Time Correction factor (TC):

$$LSTM = 15^\circ z, \quad TC = 4(\lambda - LSTM) + EoT, \quad LST = h + \frac{TC}{60} \quad (3)$$

The solar hour angle ω describes the sun's angular displacement from local noon:

$$\omega = 15^\circ(LST - 12) \quad (4)$$

Using spherical trigonometry, we derive the solar altitude α and solar azimuth γ (conforming to the convention: South=0°, West is positive):

$$\sin \alpha = \sin \phi \sin \delta + \cos \phi \cos \delta \cos \omega, \quad \alpha = \arcsin(\sin \alpha) \quad (5)$$

$$\sin \gamma = \frac{\cos \delta \sin \omega}{\cos \alpha}, \quad \cos \gamma = \frac{\sin \alpha \sin \phi - \sin \delta}{\cos \alpha \cos \phi}, \quad \gamma = \text{atan2}(\sin \gamma, \cos \gamma) \quad (6)$$

Step B: Solar Vector Construction (ENU Coordinates)

To facilitate vector operations, we transform the angular data into a unit vector $\mathbf{s}$ within the East-North-Up (ENU) coordinate system. We first convert the South-based azimuth γ to a North-based bearing A_N :

$$A_N = 180^\circ - \gamma \quad (7)$$

The solar direction unit vector $\mathbf{s} = (x_e, y_n, z_u)$ is then constructed as:

$$\mathbf{s} = \begin{bmatrix} x_e \\ y_n \\ z_u \end{bmatrix} = \begin{bmatrix} \cos \alpha \sin A_N \\ \cos \alpha \cos A_N \\ \sin \alpha \end{bmatrix}, \quad \text{where } \|\mathbf{s}\| = 1 \quad (8)$$

Step C: Facade Incidence and Irradiance Projection

For any given facade orientation $f \in \{N, S, E, W\}$, we define the surface normal vector $\mathbf{n}_f$. For example, the normals are defined as $\mathbf{n}_N = (0, 1, 0)$, $\mathbf{n}_S = (0, -1, 0)$, $\mathbf{n}_E = (1, 0, 0)$, and $\mathbf{n}_W = (-1, 0, 0)$.

The cosine of the incidence angle $c_f = \cos \theta_{i,f}$ is calculated via the dot product, clamped to zero to account for self-shading (i.e., the sun is behind the wall):

$$c_f = \max(0, \mathbf{s} \cdot \mathbf{n}_f) \quad (9)$$

Finally, the effective beam irradiance projected onto the vertical surface, $I_{beam,f}$, is derived from the Direct Normal Irradiance (DNI):

$$I_{beam,f} = DNI \cdot c_f \quad (10)$$

5 HARMONY-NetZero: Low-Latitude

5.1 Why We Use HARMONY

The problem statement notes that the traditional “solar-noon on solstices” sizing rule is a simplistic representation and that future metrics must account for change. Accordingly, we adopt a hybrid model that (i) resolves solar geometry and shading at an *hourly* time step, (ii) evaluates performance only during *academic occupied hours* via a schedule mask, and (iii) couples *shading–daylighting tradeoffs* with a lightweight thermal balance.

Compared with a purely geometric (solstice-noon) approach, the proposed model has three advantages:

- **Time-resolved realism:** shading effectiveness depends on hour-by-hour solar altitude/azimuth and facade orientation, capturing morning/evening low-angle sun and seasonal variation.
- **Operational relevance:** metrics are computed on the academic-hour mask $m(t)$, preventing annual averages from obscuring classroom comfort.

5.2 Modeling Framework

(A) Exterior shading (overhang + fins) as a transmittance reduction

$$SF_f(t) = S_f(\alpha(t), \psi(t); D_s, D_f, S_f, H_w) \in [0, 1] \quad (11)$$

Define the facade shading factor $SF_f(t)$ as the fraction of window effectively blocked by the exterior device geometry.

(B) Window solar heat gain with glazing + dynamic curtain

$$Q_{\text{sol}}(t) = A_w \text{SHGC} \left[\text{DNI}(t) \cos \theta_f(t) (1 - SF_f(t)) + k_d \text{DHI}(t) \right] \chi_c(t) \quad (12)$$

Compute transmitted solar heat gain through the window area A_w using glazing SHGC, exterior shading reduction, diffuse factor k_d , and curtain state $\chi_c(t)$.

$$\chi_c(t) = \begin{cases} \tau_{\text{sol}}, & \text{curtain closed} \\ 1, & \text{curtain open} \end{cases} \quad (13)$$

Model the curtain as a multiplicative attenuation on solar heat gain (closed vs. open).

(C) Daylighting–shading coupling via incremental lighting electricity

$$E_{\text{light,add}} = \sum_{t=1}^T m(t) P_L \left[1 - \eta_{\text{day}}(t) \right]_+ \Delta t \quad (14)$$

Add lighting energy only when daylight fraction $\eta_{\text{day}}(t)$ (driven by VLT and curtain state) falls below target; $[x]_+ = \max(x, 0)$.

(D) Lightweight thermal balance (1R–1C surrogate)

$$C \frac{T_z(t+1) - T_z(t)}{\Delta t} = \frac{T_a(t) - T_z(t)}{R} + Q_{\text{int}}(t) + Q_{\text{sol}}(t) - Q_{\text{HVAC}}(t) \quad (15)$$

Evolve zone temperature T_z using an hourly RC balance with envelope conductance $1/R$, thermal capacitance C , internal gains Q_{int} , solar gain, and HVAC extraction.

$$E_{\text{cool}} = \sum_{t=1}^T m(t) \frac{[Q_{\text{HVAC}}(t)]_+}{\text{COP}} \Delta t \quad (16)$$

Convert thermal cooling to electrical consumption using a fixed coefficient of performance (COP).

(E) Objective and feasibility constraints

$$\min J = E_{\text{cool}} + E_{\text{light,add}} \quad (17)$$

Minimize an electricity proxy that explicitly prices the “always-close-the-curtain” loophole.

$$\sum_{t=1}^T m(t) \mathbb{I}[T_z(t) \notin [T_{\min}, T_{\max}]] = 0 \quad (18)$$

Enforce zero academic-hour comfort violations under HVAC setpoint clamping.

(F) Net-zero cooling proxy with rooftop PV

$$PV_{\text{kWh}} = \eta_{\text{mod}} PR A_{\text{roof}} \sum_{t=1}^T \text{GHI}(t) \Delta t \quad (19)$$

Estimate annual rooftop PV generation from GHI using module efficiency η_{mod} and performance ratio PR .

$$A_{\text{PV,req}} = \frac{E_{\text{cool}} + E_{\text{light,add}}}{PV_{\text{kWh}}/A_{\text{roof}}} \quad (20)$$

Compute the PV area required to offset the annual electricity proxy.

5.3 Parameter Search and Selected Retrofit Package

We sample feasible retrofit designs in a bounded parameter space (glazing, overhang, fins, thermal mass, and night ventilation), simulate hourly performance, and select the feasible design.

5.3.1 Selected design variables

Table 2: Best feasible retrofit design parameters (Sungrove/Phoenix proxy)^[4]Xu2025AnhuiNZEB.

Variable	Value	Unit
SHGC	0.1823	-
VLT	0.6266	-
Window U -value U_{window}	2.9498	W/m ² K
South overhang depth D_S	1.2801	m
East fin depth $D_{f,E}$	0.7125	m
East fin spacing $S_{f,E}$	0.4823	m
West fin depth $D_{f,W}$	0.7106	m
West fin spacing $S_{f,W}$	0.5966	m
Thermal mass multiplier m_{mass}	1.4432	-
Night ventilation rate $\text{ACH}_{\text{night}}$	3.6651	-

5.3.2 Constructible retrofit bundle

- **High-performance glazing:** specify a low-SHGC double-glazed unit while maintaining mid-high VLT to reduce daytime lighting.
- **South fixed overhang:** add a continuous exterior canopy of depth $D_S \approx 1.28$ m along the south façade.
- **East/West vertical fins (brise-soleil):** install lightweight, fixed vertical fins with depths ≈ 0.71 m and the above spacings to suppress low-angle morning/afternoon direct sun.
- **Night-flush ventilation:** implement controlled night ventilation (≈ 3.67 ACH) to pre-cool the thermal mass.
- **Added thermal mass:** increase effective internal capacitance (e.g., exposed slab/PCM panels), consistent with $m_{\text{mass}} \approx 1.44$.
- **Dynamic interior roller shades:** close shades only when glare/solar thresholds are exceeded (not continuously), to balance cooling reduction and lighting penalty.

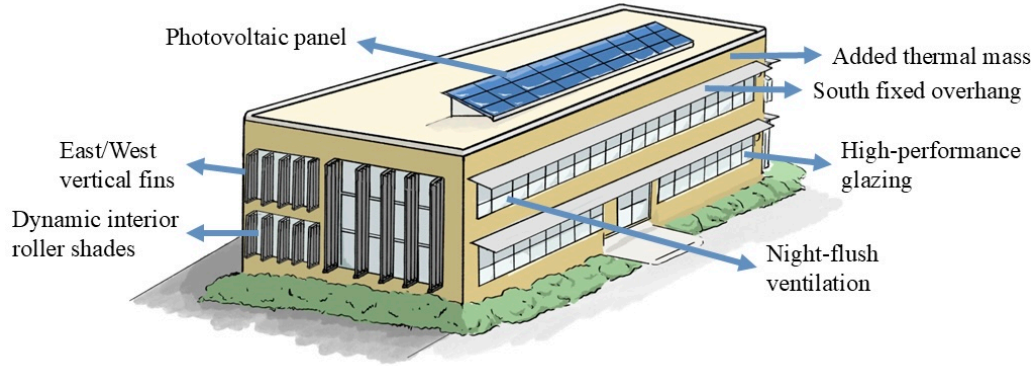


Figure 5: the simulated diagram of the buildings

5.4 Instantiated Inputs and Key Results

5.4.1 Climate and schedule evidence

We use a Phoenix, AZ hourly EPW proxy for Sungrove. The dataset exhibits extreme heat and strong direct normal irradiance, motivating SHGC reduction and exterior shading. Academic occupied hours total 2772 h (31.64% of the year), concentrated at local times 8–19, including afternoon west exposure.

5.4.2 Cooling-load reduction and electricity proxy

Table 3: Performance summary of the best retrofit (baseline* vs best feasible).

Metric	Baseline*	Best retrofit	Improvement
Academic cooling (kWh _{th})	442,885.6	248,024.5	−44.0%
Annual cooling (kWh _{th})	726,094.1	377,360.9	−48.0%
Curtain closed fraction (academic)	0.5325	0.3402	↓
Curtain closed hours (academic)	1476/2772	943/2772	↓
Incremental lighting (academic, kWh)	10,998.9	1,142.5	−89.6%
Total electricity proxy (annual, kWh)	—	72,006.6	(reported)
PV annual yield on roof (kWh)	—	638,091.3	(reported)
PV area for 100% offset (m ²)	—	162.5	(reported)

5.4.3 Constraint satisfaction

During academic hours, the best retrofit achieves zero comfort-violation hours and satisfies the glare constraint under the adopted glare proxy/threshold.

5.5 Robustness, Sensitivity, and 2040 Prediction

5.5.1 Heatwave operational stress test

$$E_{\text{acad,cool}}^{(\text{heatwave})} = 246,207.4 \text{ kWh}_{\text{th}}, \quad \text{comfort violations} = 0 \quad (21)$$

Under a heatwave operations test, the retrofit remains feasible with zero academic comfort violations and comparable academic cooling magnitude.

5.5.2 2040 warming scenarios (cooling feasibility)

$$E_{\text{acad,cool}}^{(2040,\text{low})} = 254,986.7 \text{ kWh}_{\text{th}} \quad (22)$$

Projected academic cooling under a mild 2040 warming scenario.

$$E_{\text{acad,cool}}^{(2040,\text{mid})} = 263,770.2 \text{ kWh}_{\text{th}} \quad (23)$$

Projected academic cooling under a medium 2040 warming scenario.

$$E_{\text{acad,cool}}^{(2040,\text{high})} = 272,520.5 \text{ kWh}_{\text{th}} \quad (24)$$

Projected academic cooling under a severe 2040 warming scenario (about +9% vs. baseline-year academic cooling of the best design).

5.5.3 Sensitivity protocol

We evaluate parameter sensitivity by perturbing uncertain inputs (e.g., envelope UA, internal gains, night ventilation rate, and glazing aging) and tracking feasible-set stability and ranking of designs.

5.5.4 Net-zero cooling readiness by 2040

Because Phoenix solar resource yields ≈ 638 MWh annual rooftop PV in our proxy while the optimized cooling+lighting electricity proxy is ≈ 72 MWh, only $\approx 162.5 \text{ m}^2$ (about 11% of a 1440 m^2 roof) is required for full offset in the base year. Given that the 2040 cooling increase remains within a single-digit percentage range under the tested scenarios, the net-zero cooling target is achievable with a modest PV allocation while the main feasibility risk shifts to maintaining comfort under heatwave and warming tails.

6 HARMONY-NZ-ThermalMass: High-Latitude

Requirement 2 asks for a retrofit strategy in a cold, high-latitude city (Borealis) where heating dominates and thermal mass is critical. Building upon the previous HARMONY-NetZero model, we have developed an enhanced version that couples the following aspects:

- (i) solar geometry and facade incidence
- (ii) simplified external shading geometry (overhangs and fins)
- (iii) a glare proxy with seasonally-enabled interior shades
- (iv) a one-zone thermal balance with explicit effective thermal capacitance to represent thermal mass
- (v) setpoint-based HVAC energy accounting

6.1 Model Advantages Compared to Traditional Approaches

Traditional building-energy screening for retrofits often relies on steady-state or coarse time-resolution methods (e.g., degree-day heat loss, monthly balance, or simplified glazing rules). Our model improves on these by:

- **Hourly coupling of sun, shading, and HVAC:** heating/cooling loads respond to hourly DNI/DHI and changing sun angles rather than assuming a daily or monthly average.
- **Explicit thermal mass (thermal capacitance) in dynamics:** thermal mass is not a qualitative label but a parameter (`mass_scale`) that directly changes the zone heat capacity and therefore temperature swing and HVAC demand.
- **Glare-aware yet heating-first control:** glare is enforced via a simple, computable proxy, and interior shades are *seasonally enabled* (warm months only) to avoid blocking valuable winter solar gains.
- **Feasible-set search rather than hand-tuning:** instead of picking one retrofit guess, we explore a design space and filter for feasible solutions under constraints, then rank by heating-first objectives.

6.2 Model Formulation

*Cooling energy is tracked but not the primary objective. We prioritize heating reduction subject to glare/comfort constraints, while reporting cooling as a trade-off and for future climate robustness.

(A) Inputs

We simulate hourly weather time series (EPW) spanning multiple years (2009–2023) and evaluate only *academic hours* using a schedule mask:

$$\mathcal{T}_{\text{acad}} = \{t \mid \text{academic_mask}(t) = 1\}, \quad |\mathcal{T}_{\text{acad}}| = 2772 \text{ hours.}$$

Let $T_a(t)$ be outdoor temperature, $DNI(t)$ direct normal irradiance, $DHI(t)$ diffuse horizontal irradiance. The model uses a single thermal zone with indoor temperature $T_z(t)$.

(B) External Shading

We approximate external shading by a dimensionless shading factor $SF_f(t) \in [0, 1]$ applied to the *beam* component on the corresponding facade:

$$\text{beam}_f(t) = DNI(t) c_f(t) (1 - SF_f(t)).$$

(C) Solar Heat Gain Through Windows

Let A_f be window area on facade f and $SHGC$ be the solar heat gain coefficient of the glazing system. Diffuse contribution is modeled as $DHI(t)F_{\text{sky}}$ with constant F_{sky} :

$$Q_s(t) = \sum_f A_f SHGC \left[\underbrace{DNI(t) c_f(t) (1 - SF_f(t))}_{\text{beam}} + \underbrace{DHI(t) F_{\text{sky}}}_{\text{diffuse}} \right].$$

This $Q_s(t)$ is the primary source of passive winter heating benefit but also a driver of warm-season overheating risk.

(F) Ventilation / Night Flush

Ventilation heat transfer is:

$$Q_{\text{vent}}(t) = \dot{m}(t)c_p(T_a(t) - T_z(t)), \quad \dot{m}(t) = \rho \frac{ACH(t)V}{3600}.$$

Night-flush control increases $ACH(t)$ within a night window if $T_a(t) \leq T_z(t)$ (useful mainly for warm-season heat rejection).

(G) One-Zone Energy Balance

Let UA be envelope conductance and $Q_{\text{int}}(t)$ internal gains. For time step Δt :

$$T_{\text{pred}}(t + \Delta t) = T_z(t) + \frac{\Delta t}{C_{\text{th}}} \left(Q_s^{\text{eff}}(t) + Q_{\text{int}}(t) + Q_{\text{vent}}(t) - UA(T_z(t) - T_a(t)) \right).$$

6.3 Model Results, Conclusions, and Interpretation

6.3.1 Feasible-Set Search Outcome

We sample the retrofit design space (glazing, shading geometry, thermal mass, night ventilation) and filter designs that meet:

- glare-hours $\leq G_{\text{target}}$,
- comfort violations = 0 (setpoint band maintained during academic hours),
- daylight deficit hours kept track as a secondary penalty,
- additional practical bounds (nonnegative dimensions, plausible U/SHGC/VLT ranges).

Feasible designs found: **1194**.

6.3.2 Best Design Parameters

Parameter	Value
$SHGC$	0.8488
U_{window} (W/m ² K)	0.7047
VLT	0.6288
South overhang depth $D_{\text{oh,S}}$ (m)	0.6970
East fin depth $D_{\text{fin,E}}$ (m)	0.3182
East fin spacing $S_{\text{fin,E}}$ (m)	0.4042
West fin depth $D_{\text{fin,W}}$ (m)	0.2087
West fin spacing $S_{\text{fin,W}}$ (m)	0.7075
Thermal mass scale (mass_scale)	5.9725
Night ACH (ach_night)	0.2374
Cost estimate (\$)	96,116.8

6.3.3 Key Performance Metrics

Metric	Value
Baseline glare hours (no shades)	394
Glare target hours	275.8
Best-design glare hours	233
Baseline academic heating (kWh)	49,798.8
Best-design academic heating (kWh)	40,119.2
Heating saving fraction	19.4%
Best-design academic cooling (kWh)	94,952.9
Comfort violation hours (academic)	0
Daylight deficit hours (academic)	150
Passive winter underheat hours (no heating)	3596
Passive warm overheat hours (JJA, no cooling)	1978
Winter temperature swing p95–p05 (°C)	3.26

*Comfort violation hours = 0 does not mean the building is naturally comfortable; it means our retrofit comparison does not allow comfort sacrifice.

6.3.4 Thermal Mass (Material Proxy) vs Heating

Holding geometry and operation fixed, the material/thermal-mass comparison shows monotonic improvement in heating with higher `mass_scale`:

Material proxy	<code>mass_scale</code>	E_{heat} (kWh)
Concrete	6.00	41,896.3
Stone	5.92	41,910.0
Brick	4.35	42,375.3
Gypsum	2.98	43,121.1
CLT/Wood	2.17	43,862.3

Takeaway: high thermal mass reduces heating energy by smoothing indoor temperature and retaining solar/internal gains, which is especially valuable in Borealis. (Absolute values can differ from the constrained best-design run because the comparison is a controlled “swap” study; the ranking and sensitivity are the main evidence.)

Thermal mass implementation. To realize $m_s = 5.9725$, interior thermal mass is enhanced through exposed concrete floor slabs and selected heavy interior partitions (e.g., masonry or concrete-backed finishes) in high-irradiance zones. This stores winter solar gains and releases them over several hours.

6.3.5 Passive Survivability Diagnostics

To demonstrate thermal mass benefits beyond HVAC energy, we also run “HVAC-off” diagnostics:

- Passive winter underheat hours (all winter months, no heating): 3596 h
- Passive warm overheat hours (Jun–Aug, no cooling): 1978 h
- Winter temperature swing (p95–p05): 3.26°C

Interpretation: thermal mass reduces short-term temperature volatility (low swing), improving resilience, even though extreme cold still requires heating (large underheat hours without HVAC).

6.4 Future Robustness^[2]

Under 2040 warming, E_{heat} decreases but E_{cool} rises. This suggests a staged plan:

1. Implement heating-focused envelope + thermal mass now (high return in Borealis).
2. Reserve retrofit flexibility for later: stronger warm-season shading, improved night ventilation, or high-efficiency cooling as climate warms.

We apply uniform warming and quantify the shift in loads:

Scenario	E_{heat} (kWh)	E_{cool} (kWh)	Passive Overheat (JJA, h)
2040_low ($\Delta T = 1^\circ\text{C}$)	37,585.5	101,225.1	2014
2040_mid ($\Delta T = 2^\circ\text{C}$)	35,138.8	107,522.3	2057
2040_high ($\Delta T = 3^\circ\text{C}$)	32,761.4	113,779.3	2106

Conclusion: warming reduces heating energy (beneficial for Borealis) but increases cooling burden and passive overheating risk; therefore, shading and/or ventilation strategies may need strengthening over time even in a heating-dominant city.

6.5 Core Clarifications (Avoiding Common Misinterpretations)

1. **Academic-year lens:** reported E_{heat} and E_{cool} are integrated over academic hours, not all 8760 hours. This matches the school-use focus and prevents summer vacation hours from dominating results.
2. **Baseline “no shades” is intentional:** we disable interior shades in baseline to avoid giving the baseline an operational advantage that hides glare at the cost of winter heating.
3. **Glare metric is a proxy:** glare hours are based on a conservative irradiance–transmittance proxy, not a full Daylight Glare Probability (DGP) simulation. The goal is consistent, tractable screening across thousands of candidates.
4. **Thermal mass is represented by effective heat capacity:** `mass_scale` is an equivalent parameter capturing the aggregate influence of interior structural mass and exposed surfaces; it is not a literal material thickness.

7 HARMONY-NZ-TM-Generalized: Multiple Locations

Requirement 3 asks us to *adapt* our shading/comfort/energy model beyond the two notional campuses and *discuss design considerations* for other locations, including cases at similar latitudes but with different heating/cooling needs. We address this by keeping a single physics-based core model (hourly heat balance + dynamic operable shades + daylight/lighting coupling) and adding a **location configuration layer** that changes only:

- (i) climate data (EPW)
- (ii) constraints/weights reflecting local priorities
- (iii) feasible design bounds

We then run the same search-and-screen workflow for each city and compare results with two groups:

Phoenix, San Diego, Atlanta (warm climates)

Fairbanks, Reykjavik, Yakutsk (cold climates)

7.1 Advantages Over Traditional Models

1. **Fast enough for multi-city optimization:** We use an hourly, reduced-order thermal model with explicit solar/shading terms, so we can evaluate thousands of designs per city and still perform robust scenario screening.
2. **Human-comfort realism through dynamic operable shades:** Instead of treating shading as a fixed geometry only, we model *glare-triggered* shade deployment that directly trades off glare, daylight availability, and lighting energy.

7.2 Model Formulas and Description

(A) Core hourly thermal balance

For each hour t (time step Δt), the zone temperature T_t follows a lumped heat-capacitance balance:

$$T_{t+1} = T_t + \frac{\Delta t}{C} \left[\frac{T_t^{\text{out}} - T_t}{R} + Q_t^{\text{sol}} + Q_t^{\text{int}} + Q_t^{\text{vent}} - Q_t^{\text{HVAC}} \right]. \quad (25)$$

Here R, C represent the effective envelope resistance and thermal capacitance (scaled by `mass_scale`). Q_t^{int} comes from occupancy/lighting/plug gains during academic hours. Q_t^{vent} is driven by night ventilation (`ach_night`) when enabled. The HVAC load Q_t^{HVAC} is the minimum heating/cooling power required to keep T_t within comfort bounds (e.g., $[T_{\min}, T_{\max}]$ for cold climates). The seasonal energy totals are:

$$E_{\text{cool}} = \sum_t \max(Q_t^{\text{HVAC}}, 0) \frac{\Delta t}{3.6 \times 10^6}, \quad E_{\text{heat}} = \sum_t \max(-Q_t^{\text{HVAC}}, 0) \frac{\Delta t}{3.6 \times 10^6}, \quad (26)$$

(B) Objective function and constraints

We search a retrofit design vector

$$x = \{\text{SHGC}, U, \text{VLT}, D_{\text{overhang}}^S, \text{light_shelf}, D_{\text{fin}}^E, S_{\text{fin}}^E, D_{\text{fin}}^W, S_{\text{fin}}^W, \text{mass_scale}, \text{ach_night}\}$$

with bounds tuned per location.

Among feasible designs, we minimize a composite score (lower is better):

$$J(x) = w_c E_{\text{cool}} + w_h E_{\text{heat}} + w_t E_{\text{light,add}} + w_d H_{\text{def}}^{\text{ref}} + w_s H_{\text{shade}} + w_{\$} \text{Cost} + w_g \text{GE} \\ + w_q \left(\max \left(0, \frac{H_{\text{def}}^{\text{ref}}}{H_{\text{acad}}} - f_{\text{soft}}^{\text{def}} \right) \right)^2 H_{\text{acad}}. \quad (27)$$

7.3 Model Results, Conclusions, and Explanations

7.3.1 Summary tables

Table 4 reports feasibility counts and the selected best feasible/robust outcomes for each city. (All cities achieved zero glare hours due to the operable-shade control and our glare constraints; therefore shade hours and daylight deficit become key differentiators.)

Table 4: Requirement 3 outputs (best feasible and best robust). Energies are in kWh.

City	Feas.	Robust	E_{cool}	E_{heat}	DefHrs	ShadeHrs	$E_{\text{light,add}}$	Cost(\$)	WorstFutureCool	ReqPVarea(m ²)
Phoenix	283	80	94310.45	0.00	1	1	28.80	162337	108082.3	194.6
San Diego	88	80	104239.22	0.00	368	368	8478.72	138747	126766.4	205.4
Atlanta	233	80	68488.91	218.50	72	72	2073.60	114776	88251.0	144.6
Fairbanks	249	80	46158.39	54573.25	167	167	3847.68	128651	34871.6	240.3
Reykjavik	270	80	10848.80	9215.03	135	135	3110.40	126387	18163.0	103.6
Yakutsk	172	80	48014.56	82661.59	197	197	5106.24	126642	36383.6	234.4

Table 5: Best feasible design parameters by city.

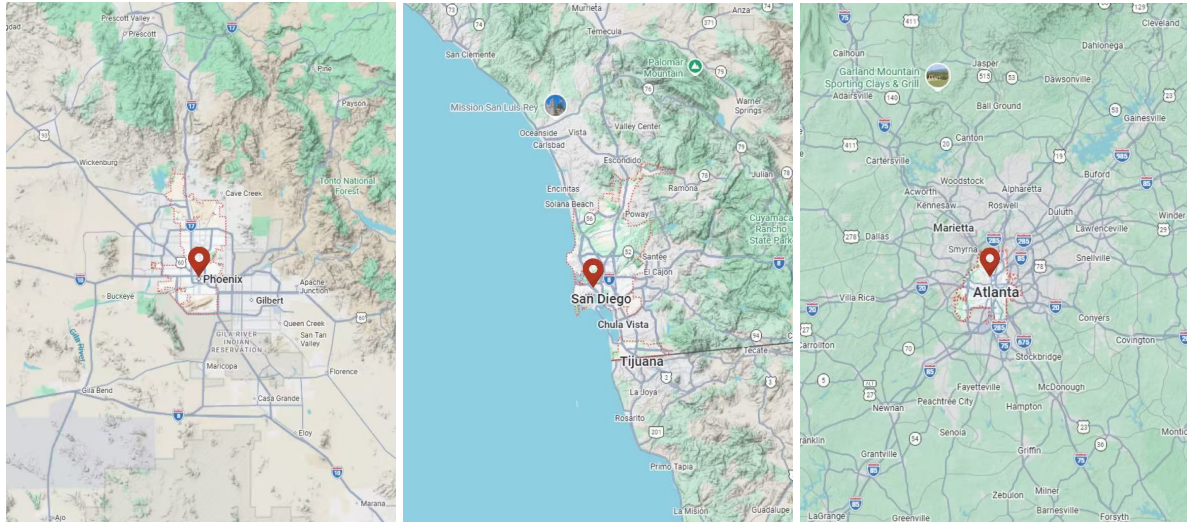
City	SHGC	U	VLT	D_{ov}^S	Shelf	D_{fin}^E	S_{fin}^E	D_{fin}^W	S_{fin}^W	mass_scale	ach_night
Phoenix	0.153	2.258	0.402	1.990	1	0.283	0.677	0.824	1.402	1.561	0.940
San Diego	0.253	2.658	0.502	1.194	1	0.188	0.777	0.549	1.502	1.401	0.618
Atlanta	0.162	2.469	0.491	1.028	0	0.848	1.414	0.372	0.586	3.595	0.343
Fairbanks	0.595	1.339	0.569	0.856	1	0.176	1.297	0.476	1.604	4.235	0.431
Reykjavik	0.330	1.613	0.611	0.767	1	0.371	1.819	0.291	1.366	5.847	0.440
Yakutsk	0.694	1.139	0.572	0.856	1	0.141	1.625	0.381	1.976	5.577	0.380

7.3.2 Net-zero proxy

All cities achieve 100% academic-hour offset under our proxy assumptions because the required PV areas (103–240 m²) are far smaller than the available roof area (60 m × 24 m). We treat this as a *comparative* metric, not a promise of detailed electrical design.

7.4 Core Clarifications

1. **Refined daylight deficit is deliberate:** High-latitude winters are dark by nature; counting all “low-VLT” hours would incorrectly label every design as unusable. We count deficit only when outside GHI is above a threshold.
2. **Robust design is not the same as feasible optimum:** Robust screening intentionally prioritizes worst-case cooling under heatwaves and 2040 deltas, so it may trade some heating savings for summer resilience (seen in Fairbanks and Yakutsk).



(a) Phoenix

(b) San Diego

(c) Atlanta

Figure 6: the topography of different cities in similar low latitudes



Figure 7: the topography of different cities in similar high latitudes

8 HARMONY's Application

Requirement 4 requests a *new* Student Union building design that relies heavily on passive solar shading (not mechanical cooling), performs well throughout the day (not only at solar noon), supports non-cardinal window orientations, accommodates diverse window geometries, and remains effective under future climate conditions. Our solution meets these requirements by coupling:

- **New-build degrees of freedom (DOF):** building rotation and facade-specific window-to-wall ratios (WWR),
- **Passive envelope/shading DOF:** glazing properties, overhangs, vertical fins, light shelf, thermal mass scaling, and night ventilation,
- **Scenario-based robustness:** present climate, operational heatwave stress, 2040 climate deltas, and outage survivability,
- **Sensitivity (PRCC) analysis:** interpretable global importance rankings for key inputs.

8.1 HARMONY's Advantages

1. **Whole-day, whole-year evaluation:** We simulate at hourly resolution to capture morning/afternoon low-sun-angle penalties and realistic daily shading needs.
2. **Rotation + WWR co-optimization:** Unlike fixed-orientation templates, we explicitly treat rotation and facade-specific WWR as decision variables.
3. **Baseline-normalized glare control:** For each rotation+WWR building instance, we compute a baseline glare reference and enforce improvement, preventing false “wins” caused solely by shrinking windows.
4. **Future-proofing by design:** We rank solutions by worst-case future performance (2040 deltas) and additional stress tests (heatwave operations, outage survivability).
5. **Passive-first validation:** We explicitly test HVAC-off free-float behavior and a minimal “backup HVAC” mode to quantify how far passive measures can go under extreme conditions.
6. **Interpretability:** PRCC-based sensitivity provides actionable guidance (which variables most strongly drive energy/robustness outcomes).

8.2 Model Formulation

(A) Decision variables

We define a design vector

$$\mathbf{x} = (\theta, \text{WWR}_S, \text{WWR}_N, \text{WWR}_{EW}, \underbrace{\text{SHGC}, U, \text{VLT}, D_{ovh}, \dots}_{\text{passive/retrofit DOF}}),$$

where:

- θ is the building rotation angle (degrees),
- $\text{WWR}_S, \text{WWR}_N, \text{WWR}_{EW}$ are facade WWRs (East/West share one value),
- Passive/retrofit DOF include glazing (SHGC, U , VLT), south overhang depth D_{ovh} , east/west fin depths and spacings, light shelf switch, thermal mass scale, and night ventilation rate.

(B) Geometry mapping from WWR to window areas

For a fixed building size (length L , width W , total height H), facade wall areas are:

$$A_{NS} = L H, \quad A_{EW} = W H.$$

Window areas are computed by:

$$\begin{aligned} A_{win,S} &= \text{WWR}_S A_{NS}, & A_{win,N} &= \text{WWR}_N A_{NS}, \\ A_{win,E} &= \text{WWR}_{EW} A_{EW}, & A_{win,W} &= \text{WWR}_{EW} A_{EW}. \end{aligned} \quad (28)$$

Rotation θ modifies each facade azimuth, thereby changing incident solar angles and shading effectiveness over the day.

(C) Operating schedule and internal gains

We build an operating mask $m(t) \in \{0, 1\}$ (weekday: 7–23, weekend: 9–23). Internal gains are:

$$Q_{int}(t) = m(t) A_{floor} g(t), \quad (29)$$

where A_{floor} is total floor area, and $g(t)$ is a piecewise gain density (base vs. peak windows). In our implementation:

$$g(t) = \begin{cases} g_{peak}, & t \in \text{peak periods} \\ \max(g_{base}, 0.75 g_{peak}), & t \in \text{weekday shoulder (15–17)} \\ g_{base}, & \text{otherwise during open hours} \\ 0, & \text{closed} \end{cases}$$

with $g_{base} = 8 \text{ W/m}^2$ and $g_{peak} = 16 \text{ W/m}^2$.

8.3 Results, Conclusions, and Explanations

8.3.1 Search outcome

Our Latin Hypercube sampling explored 600 candidates. In this run:

Feasible designs found: 600/600, **Robust designs found:** 600/600.

This indicates that within our chosen bounds, all sampled designs satisfied the (stringent) glare/comfort constraints; optimization therefore reduces to *ranking* by energy and cost, and robustness reduces to *ranking* by worst-case future energy and survivability metrics.

8.3.2 Best base-feasible design (includes new-build DOF)

The best base-feasible design is:

- Rotation: $\theta = -5.5^\circ$
- WWR: $\text{WWR}_S = 0.23$, $\text{WWR}_N = 0.30$, $\text{WWR}_{EW} = 0.07$
- Passive/retrofit:

$$\text{SHGC} = 0.1806, \quad U = 1.357 \text{ W}/(\text{m}^2\text{K}), \quad \text{VLT} = 0.3896, \quad D_{ovh} = 0.055 \text{ m},$$

light shelf on, east fins ($D = 0.463 \text{ m}$, $S = 0.190 \text{ m}$), west fins ($D = 0.255 \text{ m}$, $S = 0.447 \text{ m}$), mass scale = 2.105, night ventilation ACH = 1.234.

Performance:

$$E_{cool} = 517\,270 \text{ kWh}, \quad E_{light} = 0 \text{ kWh}, \quad H_{glare} = 0, \quad H_{viol} = 0, \\ E_{elec} = 147\,791.43 \text{ kWh}, \quad C_S = 96\,103.36.$$

Interpretation. The optimal solution suppresses solar gains primarily through a low SHGC glazing and strong east/west vertical fins (critical for low-angle sun), while keeping east/west WWR very small. Moderate VLT plus a light shelf maintains daylighting, yielding $E_{light} \approx 0$. Increased thermal mass and night ventilation provide passive damping of temperature swings.

8.3.3 Best robust design (base + heatwave ops + 2040 deltas)

In this run, the best robust design equals the best base-feasible design, but with additional robust metrics:

$$E_{2040}^{worst}(cool) = 594\,974.94 \text{ kWh}, \quad E_{2040}^{worst}(elec) = 169\,992.84 \text{ kWh}.$$

This is a **15.0%** increase relative to base electrical demand:

$$\frac{169992.84 - 147791.43}{147791.43} \times 100\% = 15.0\%.$$

Outage survivability:

$$T_{max}^{outage} = 55.11^\circ\text{C}, \quad H_{T>32^\circ\text{C}}^{outage} = 39 \text{ hours}.$$

Interpretation. Even with excellent shading and low SHGC, a hot future climate increases cooling electricity by about 15%. The outage metrics are intentionally conservative: without active cooling, indoor temperatures can reach unsafe levels during extreme events. This supports a design recommendation of *passive-first + contingency*: shading and envelope measures reduce baseline demand, while emergency measures (cool refuge areas, backup power, or limited mechanical support) address rare but critical conditions.

8.4 Passive-first comparison (HVAC OFF) and backup HVAC

We evaluated the robust-best design under two additional modes.

HVAC OFF (fully passive free-float). During operating hours:

$$T_{peak} = 64.324^\circ\text{C}, \quad T_{min} = 0.969^\circ\text{C}, \quad H_{outside} = 5195, \quad \text{fraction outside band} = 0.923.$$

Threshold exceedance hours:

$$H_{T>26^\circ\text{C}} = 4529, \quad H_{T>28^\circ\text{C}} = 4313, \quad H_{T>30^\circ\text{C}} = 4058.$$

Degree-hour severity:

$$\text{overheat deg-h} = 73647.424, \quad \text{underheat deg-h} = 4350.570.$$

Backup HVAC (relaxed setpoints). With setpoints $T_{cool,set} = 28.5^\circ\text{C}$ and $T_{heat,set} = 18^\circ\text{C}$:

$$E_{cool} = 408\,701.62 \text{ kWh}, \quad E_{heat} = 950.64 \text{ kWh}, \quad E_{elec} = 116\,771.89 \text{ kWh}.$$

This reduces electrical cooling+lighting demand by **21.0%** relative to the base-feasible full-comfort case:

$$\frac{147791.43 - 116771.89}{147791.43} \times 100\% \approx 21.0\%.$$

However, comfort violations remain large ($H_{viol} = 5104$) because this mode is explicitly *not* full comfort; it is a survivability/energy-saver mode with permissive setpoints.

Conclusion for requirement 4 Passive solar shading and envelope design can drastically reduce solar heat gains and make the building efficient under normal operation, but **purely passive operation is not sufficient** to maintain comfort in this hot climate with high internal gains. A **passive-first** design should therefore be paired with a **minimal backup strategy** (limited HVAC or emergency cooling) for resilience.

8.5 Core Clarifications

1. **Why 600/600 feasible?** Our sampled bounds already enforce reasonable passive design (low-to-moderate WWR, strong shading options). Within that region, glare and comfort constraints were satisfied for all samples; thus optimization focuses on ranking and robustness metrics, not feasibility filtering.
2. $E_{light} = 0$ **does not mean “no lighting.”** It means *additional* lighting energy beyond baseline is approximately zero because daylighting is sufficient under the adopted daylight control assumptions.
3. **Small south overhang depth is not an error.** The model can trade off among shading mechanisms; in this best design, low SHGC + fins + small WWR (especially E/W) dominate, reducing the need for a large south overhang.
4. **HVAC-off extremes are expected.** The HVAC-off mode is a stress test. The very high free-float peak temperature confirms that passive shading alone cannot guarantee comfort during extreme heat and occupied hours; it supports the need for contingency planning.

Table 6: Key performance comparison for the selected Student Union design.

Metric	Base-feasible	Robust worst-2040	HVAC OFF (passive)	Backup HVAC
Rotation (deg)	-5.5	-5.5	-5.5	-5.5
WWR_S	0.23	0.23	0.23	0.23
WWR_N	0.30	0.30	0.30	0.30
WWR_{EW}	0.07	0.07	0.07	0.07
E_{cool} (kWh)	517270.00	594974.94	—	408701.62
E_{elec} (kWh)	147791.43	169992.84	—	116771.89
E_{light} (kWh)	0.00	—	—	0.00
H_{glare} (h)	0	0	—	0
H_{viol} (h)	0	0	5195	5104
Outage T_{max} ($^\circ\text{C}$)	—	55.11	—	—
Outage hours T	—	39	—	—
Cost (\$)	96103.36	96103.36	—	—

9 Weaknesses and Improvements

9.1 Weaknesses

1. The operation of real campuses under extreme weather conditions faces additional constraints, posing extrapolation risks to policy benefits.
2. Thermal mass may underestimate (or overestimate) the peak shaving and valley filling capabilities of certain materials during daily cycles.

9.2 Improvements

1. Layered, system-level passive envelope (exterior + glazing + interior) We model the student union façade as a cascaded passive-control system: exterior geometry (overhang + fins) attenuates direct beam, adaptive glazing modulates transmitted solar heat by season, and an interior micro-louver layer trims residual peaks while preserving useful daylight.
2. PV canopy as an active boundary-condition modifier (shade + generate) The PV canopy coverage is embedded as an additional beam-blocking mechanism on the south façade, i.e., it directly changes incident solar gains while simultaneously producing electricity.
3. Seasonally adaptive solar heat gain coefficient (SHGC) We represent “smart” glazing using two SHGC states ($SHGC_{hot}$, $SHGC_{mild}$) to explicitly model seasonal switching, rather than assuming a constant transmittance year-round.

References

- [1] Myles R. Allen, William F. Lamb, Asta Malinauskaite, and Joeri Rogelj. Net Zero: Science, Origins, and Implications. *Annual Review of Environment and Resources*, 47:849–887, 2022. doi:10.1146/annurev-environ-112320-105050.
- [2] Nathaniel Tarshish, Nadir Jeevanjee, and Inez Fung. Cooling after net zero. EarthArXiv preprint, n.d.
- [3] Laura Aelenei, M. Graça Carvalho, Holger Hass, Helena Gonçalves, and Rui Patricio. Passive approaches for cooling in NZEB design. In *CISBAT 2011* (Lausanne, Sept 14–16), 2011.
- [4] Chiemeire Ume and Halil Zafer Alibaba. Optimization of Passive Solar Design Strategies in Hot Climate Cyprus. *International Journal of Civil and Structural Engineering Research*, 6(2):140–148, 2018–2019.
- [5] Bruce Haglund and Kurt Rathmann. *Thermal Mass in Passive Solar and Energy-Conserving Buildings* (Vital Signs Curriculum Materials Project). University of Idaho, 1995.
- [6] Jun Xu, Yu Gao, and Lizhong Yang. Study on Technology Integration and Application Demonstration for Near-Zero-Energy Buildings in Central and Southern Anhui. *Sustainability*, 17:6535, 2025. doi:10.3390/su17146535.

Letter to Sungrove University

To reduce academic-year cooling demand and mitigate glare in instructional spaces, we recommend institutionalizing a *passive shading bundle*—**glazing selection + exterior shading geometry + seasonal/time scheduling**—and evaluating it under **hourly** solar conditions during **occupied teaching periods**. This produces auditable key performance indicators (KPIs) for both retrofits and new construction and enables a 2040 / heat-wave stress test.

A. Retrofit pathway (Academic Hall North and similar buildings)

1. **Set KPI thresholds** for occupied-hours cooling (E_{cool}), glare-hours, and comfort exceedance (e.g., degree-hours above an upper comfort limit).
2. **Screen feasible design bundles** with an hourly model: glazing (SHGC, U , VLT), south overhang depth, east/west vertical fins (low-angle sun and glare control), and—when needed—thermal mass and night ventilation for heat-wave resilience.
3. **Lock a constructible specification** by matching selected parameters to commercially available glazing and shading products using manufacturer datasheets.

Retrofit recommendation: SHGC = 0.18, $U_{\text{window}} = 1.97$, VLT = 0.43, $D_{\text{oh}} = 1.3$ m, $D_{\text{fin,E/W}} = 0.195/0.207$ m, $S_{\text{fin,E/W}} = 0.207/0.468$ m, mass_scale = 2.10, ach_night = 5.54.

B. New-building pathway (Student Union prototype)

1. **Commit early** to orientation/rotation, façade-specific window-to-wall ratio (WWR), and glazing–shading compatibility (avoid late-stage add-ons).
2. **Adopt a two-scenario acceptance rule:** (i) current climate (occupied hours) and (ii) 2040 / heat-wave stress test (e.g., $T_a^{(2040)}(t) = T_a(t) + \Delta T$; optional irradiance scaling).

New-build recommendation : SHGC = 0.18, $U_{\text{window}} = 1.36$, VLT = 0.39, $D_{\text{oh}} = 0.055$ m, $D_{\text{fin,E/W}} = 0.46/0.45$ m, $S_{\text{fin,E/W}} = 0.19/0.45$ m, light_shelf = 1, mass_scale = 2.104, ach_night = 1.23.

C. Implementation and verification

1. Establish a campus **component library** (approved glazing, shading assemblies, and materials).
2. Run **screening + robustness** to generate a feasible list and a robust list for replication.
3. Deliver **pilot projects** (1–2 buildings) and verify performance with minimal sensing (indoor temperature, illuminance, and energy metering) using the same KPI definitions.
4. Codify validated parameter ranges into a **campus passive-shading standard** for procurement and future design.

We would be pleased to support KPI finalization, product matching, and the pilot verification plan.

Sincerely,
Team 2604548

Report on Use of AI

To maintain academic integrity and transparency, we document our limited use of AI tools. AI was used only as an auxiliary checker (language proofreading and technical verification). All modeling choices, parameter selection, simulations/optimizations, data handling, and final writing were completed and verified by the team. No results, figures, or conclusions in this paper were generated solely by AI, and no AI output was adopted without independent verification.

1. DeepSeek(V3 version; accessed 2.2)

Query: *"In our MCM/ICM modeling, we wrote the strengths of the model in the model section, as we felt this could serve as the reason for choosing that model. We planned to discuss the weaknesses separately after explaining all the models. However, we noticed that previous O-award-winning papers typically placed both strengths and weaknesses together in the later part of the paper. What would be the best approach?"*

Output: The structural arrangement issue you encountered in your MCM/ICM paper is quite typical and reflects logical considerations in organizing the paper. Your approach has its merits, but the pattern commonly seen in O-award-winning papers does align better with MCM/ICM judging preferences. Let's analyze this and provide specific recommendations:

Comparison and Trade-offs of the Two Approaches

Your approach (strengths in the model section, weaknesses concentrated later):

Advantages: Logically natural—when introducing a model, it makes sense to explain “why it is suitable for this problem,” which improves readability and flow.

Disadvantages: Strengths are scattered, making it harder for judges to evaluate them quickly; separating strengths and weaknesses may reduce clear contrast.

Common practice in O-award-winning papers:

Advantages: More systematic, easier for judges to compare models side by side; demonstrates critical thinking; aligns with academic conventions.

Disadvantages: If not handled well, the model section may appear thin or lack sufficient justification for model selection.

2. DeepSeek(V3 version; accessed 2.1)

Query: *"I hope to win the O Award in MCM/ICM. Should the specific data calculated by the model be explained along with the model or placed later in sections like model robustness testing? Currently, I have future prediction data from the model to demonstrate its reliability."*

Output: In the MCM/ICM, aiming for the Outstanding Winner (O Award), the clarity of the paper's structure and logic is crucial. Regarding your question, the specific data calculated by the model should be reasonably distributed across different sections of the paper based on its purpose and presentation goals, rather than being concentrated entirely in one place. Here are detailed suggestions:

1. Core Data in the Model Development and Solution Section

Purpose: To showcase the model's application results and direct outputs, enabling judges to immediately understand the model's predictive or solving capabilities.

Approach: In the model development and solution section, gradually present key data (e.g., predicted values, optimization results, simulation outputs) alongside the explanation of the model. For example, if your model predicts future trends, use charts or tables here to display this data and explain its implications and practical significance.

Reasoning: This enriches the core content of the paper, allowing judges to visually assess the model's effectiveness. Combining data explanation with the model highlights its innovation and practicality.

2. Validation Data in the Model Testing Section (e.g., Robustness, Sensitivity Analysis)

Purpose: To demonstrate the model's reliability and stability, enhancing persuasiveness.

Approach: In the model testing section (typically including robustness analysis, sensitivity analysis, error analysis, etc.), present the data used for validation. For example, if you use future prediction data to prove the model's reliability (such as comparing it with historical data or calculating errors), provide a detailed analysis here and reference the relevant data.

Reasoning: Concentrating validation data allows for a systematic evaluation of the model's performance, reflecting a rigorous scientific approach. This section effectively addresses potential doubts and adds depth to the paper.

3. Addressing Your Specific Situation

You mentioned "having future prediction data to demonstrate the model's reliability," which suggests two potential uses for this data:

As model outputs: Present future prediction data in the model solution section as core results of the model.

As validation tools: In the model testing section, use this prediction data (e.g., comparing it with actual data or performing statistical tests) to argue for the model's reliability. If actual future data is unavailable, methods like cross-validation or simulated scenarios can be used.

Recommendation: Place the prediction data itself in the model solution section, while the reliability analysis based on this data (e.g., error analysis, stability testing) should go in the model testing section. This highlights the results while also showcasing the validation process.

4. Optimizing the Paper Structure

Abstract: Briefly mention key data and conclusions about model reliability.

Model Development and Solution: Explain the model steps in detail and embed core data (visualized with charts or tables).

Model Testing: Include robustness, sensitivity, etc., presenting validation data and analysis.

Conclusions and Extensions: Summarize data results and model advantages.

Appendix: Place extensive raw data or code in the appendix, but reference key parts in the main text.

5. Key Points for the O Award

Logical Coherence: Data presentation should align with the paper's main narrative, avoiding abruptness.

Highlighted Results: Present data clearly (e.g., using charts) and avoid lengthy tables.

Thorough Validation: Prove model reliability through multi-faceted testing (e.g., robustness).

Innovation: Data interpretation should reflect the model's unique insights.

In summary, strategically distributing data across the model solution and model testing sections will ensure your paper is both rich in results and rigorous in validation, maximizing your chances of winning an award. Wishing you outstanding success in the MCM/ICM!

3. DeepSeek(V3 version; accessed 2.1)

Query: *"What term should be used in mathematical modeling to describe a model as highly applicable, meaning it can reflect real-world performance under various conditions?"*

Output: In the field of mathematical modeling, the most appropriate and commonly used core terms to describe a model with wide applicability, strong robustness, and the ability to accurately reflect real-world performance under different conditions are as follows:

1. Primary Core Terms

Robustness

The most classic and professional term. Derived from the English word Robustness.

Meaning: Refers to a model's insensitivity to data disturbances, changes in assumptions, minor parameter variations, or input noise. A robust model maintains stable and reliable performance and conclusions even under non-ideal or unexpected conditions.

Usage Scenario: Widely used in control theory, statistics, optimization, and machine learning. For example, "The model exhibits strong robustness, maintaining stable prediction results even with a 20% error in parameter estimation."

Stability / Robustness (in statistics and economics)

A common Chinese translation of Robustness, with essentially the same meaning as "robustness," sometimes emphasizing the reliability of statistical inference. For example, "We conducted robustness tests to verify whether the core conclusions hold under different model specifications."

Generalization Capacity

Meaning: Specifically emphasizes a model's ability to perform well on unseen data or in new scenarios. A model with strong generalization capacity not only fits the training data well but also accurately predicts or describes unknown situations.

Usage Scenario: This is the gold standard in the field of machine learning. For example, "The neural network model performed excellently on the test set, demonstrating strong generalization capacity."

2. Other Highly Relevant High-Quality Terms

Versatility / Universality

Meaning: Indicates that the model's foundational theory or structure has broad applicability, enabling it to describe or solve a class of problems rather than a specific case.

Usage Scenario: Describes the model framework itself as having universal significance. For example, "The proposed dynamic framework exhibits versatility and can be applied to modeling various similar ecosystems."

Reliability

Meaning: Emphasizes the trustworthiness and consistency of the model's output results. Reliable results can be obtained under different conditions.

Usage Scenario: Focuses on the credibility of results. For example, “Verified through multiple sets of controlled experiments, the model’s predictions exhibit high reliability.”

Adaptability / Portability

Meaning: Refers to the model’s ability to be slightly adjusted or directly applied to new environments or problems.

Usage Scenario: Emphasizes the breadth of the model’s application. For example, “The risk assessment model demonstrates strong adaptability and has been successfully ported to three different urban transportation systems.”

3. How to Choose and Use

You can select terms based on the model’s focus:

Emphasize anti-interference and stability: Use robustness/stability.

Emphasize predicting the unknown or the field of machine learning: Use generalization capacity.

Emphasize a broadly applicable theoretical framework: Use versatility/universality.

Emphasize trustworthy results: Use reliability.

Emphasize cross-scenario application: Use adaptability/portability.

In papers or reports, a classic way to phrase it is to combine terms:

“This model not only fits the training data well but also demonstrates strong generalization capacity and robustness. Tested under various parameter settings and noise disturbances, its core output remains stable and highly reliable. This indicates that the model holds versatile value and is applicable to a wider range of practical conditions.”

Summary of limitations. We did not use AI to (i) create or fabricate data, (ii) generate final figures/tables/results, or (iii) produce unverified technical arguments. AI served only as a proofreading and verification aid; all final decisions and content are attributable to the team.